SERVICE-ORIENTED COMPUTING

FINAL PROJECT DOCUMENTATION

Student–Course Registration System

A microservice-based application built with NestJS, React and MongoDB Atlas

University	ESLSCA University
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1. Problem Definition

Universities deal every semester with the same recurring administrative problem: keeping track of which students are registered in which courses, while also maintaining clean and up-to-date data about the students themselves and the courses being offered. When all this information is bundled into a single monolithic application, several issues appear:

- A single point of failure — if one feature crashes, the whole registration system becomes unusable.
- Difficulty in scaling — peak registration weeks can overload the system while the rest of the application remains idle.
- Slow development cycles — every change forces a rebuild and redeployment of the entire system.
- Tight coupling between teams — developers cannot work independently on the student, course or registration features.

This project addresses these issues by re-engineering a student–Course Registration System using a service-oriented, microservices architecture. Each business capability lives in its own independent service with its own data, exposes its own REST API, and communicates with the others through a centralized API Gateway. The result is a system that is easier to scale, easier to test, and easier to evolve over time.

2. Project Scope

The student–Course Registration System covers the three core capabilities needed by any university registration workflow:

In scope

- Managing students — create, list, view, update and delete student profiles.
- Managing courses — create, list, view, update and delete the courses offered by the institution.
- Managing registrations — register a student in a course, list registrations (overall, by student, or by course), update or remove a registration.
- A React-based dashboard that lets staff manage all three areas from a single interface.
- An API Gateway that exposes a unified entry point for the frontend and forwards traffic to the correct microservice.
- Independent MongoDB Atlas storage per service, following the database-per-service pattern.

Out of scope

- Authentication and authorization (login, roles, JWT) — left as a future extension.
- Payment processing or financial holds on registration.
- Grade management and transcripts.
- Real-time notifications (email/SMS) on registration events.

3. Tools & Technologies

The project is built on a modern JavaScript / TypeScript stack chosen to match the requirements of the rubric and the standards expected of a production microservices system.

Layer	Technology	Version	Purpose
Backend	NestJS	11.x	Microservices & API Gateway framework
Language	TypeScript	5.7	Strongly typed server code
ORM / ODM	Mongoose	8.x	MongoDB modelling for NestJS
Database	MongoDB Atlas	7.x	Cloud-hosted NoSQL document store
Frontend	React	19.x	Single-page application
Routing (frontend)	React Router	7.x	Client-side navigation
HTTP client	Axios	1.x	REST calls (frontend & gateway)
API testing	Postman	—	Manual endpoint testing
Version control	Git + GitHub	—	Source control & collaboration
IDE	VS Code	—	Editor / debugger

Why this stack?

• NestJS is built on top of Express but adds first-class TypeScript, dependency injection, decorators and a modular structure — the same building blocks taught in Lab 9.

• MongoDB Atlas removes the operational burden of self-hosting a database and lets each microservice own its own database in the cloud.

• React is the de-facto industry standard for component-based UIs and pairs naturally with REST APIs through Axios.

4. System Architecture

The system follows a classic microservices architecture with a single API Gateway sitting between the frontend and the backend services. Each microservice is fully independent: it has its own NestJS application, its own module/controller/service layers, and its own MongoDB collection.

High-level architecture

```
+----------------------+
|   React Frontend     |
|  (localhost:3000 dev)|
+----------+-----------+
           |
           |  HTTP / JSON  (Axios)
           v
+----------------------+
|     API Gateway      |
|   NestJS — port 3000 |
+----+----+----+-------+
     |    |    |
     |    |    +------------------> Registration Service (3003) -> MongoDB
     |    +----------------------> Course Service       (3002) -> MongoDB
     +---------------------------> Student Service      (3001) -> MongoDB

                  MongoDB Atlas (cloud)
        students | courses | registrations  collections
```

Key architectural decisions

- **One responsibility per service —** Student, Course and Registration are split into three independent NestJS applications. None of them can read another service's data directly.
- **Database per service —** Each service owns its collection and never reaches into another service's data layer. Cross-service data is fetched through HTTP calls between services (and through the gateway from the client side).
- **API Gateway as the single front door —** The React client talks to one URL (localhost:3000). The gateway hides which downstream service handles each route, so services can be moved or scaled without touching the frontend.
- **Separation of model, service and controller —** Every NestJS service uses a *.schema.ts file for the Mongoose schema, a *.service.ts file for business logic and a *.controller.ts file for HTTP routing, all wired together by a *.module.ts file.
- **Stateless HTTP communication —** All inter-service traffic uses synchronous REST calls over HTTP. This keeps the system simple, observable and easy to test with Postman.

Request lifecycle

A typical request, e.g. "create a new student", flows through the system as follows:

- The React form calls POST http://localhost:3000/students with a JSON body.
- The API Gateway receives the call in GatewayController and delegates to GatewayService.forwardToStudent().
- GatewayService issues an Axios POST to http://localhost:3001/students.
- The Student Service's StudentController receives the request, hands it to StudentService.create(), which persists a new document via Mongoose.
- The saved document is returned all the way back up the chain to the React UI.

5. Microservices

The system is composed of three microservices, each implemented as an independent NestJS application. The number of services matches the number of team members, as required by the project rubric.

5.1 Student Service

Responsibility

Owns everything about students: identity, contact information, department and academic level. It exposes a CRUD API at /students and persists data in the students MongoDB collection.

Port

localhost:3001

Folder structure

```
student-service/
└── src/
    ├── main.ts                  # boots NestJS on port 3001
    ├── app.module.ts            # connects to MongoDB, imports StudentModule
    └── student/
        ├── student.schema.ts    # Mongoose schema
        ├── student.controller.ts# HTTP endpoints
        ├── student.service.ts   # business logic
        └── student.module.ts    # wires everything together
```

Schema (student.schema.ts)

```
@Schema()
export class Student {
  @Prop({ required: true })             name!: string;
  @Prop({ required: true, unique: true }) email!: string;
  @Prop({ required: true })             department!: string;
  @Prop({ required: true })             level!: string;
}
```

Endpoints

Method	Path	Description
POST	/students	Create a new student
GET	/students	List all students
GET	/students/:id	Get one student by Mongo _id
PUT	/students/:id	Update a student
DELETE	/students/:id	Delete a student

5.2 Course Service

Responsibility

Owns the catalogue of courses offered by the university. Each course carries an identifying code, a name, the instructor, credit hours, the department it belongs to, capacity and schedule.

Port

localhost:3002

Folder structure

```
course-service/
└── src/
    ├── main.ts                # boots NestJS on port 3002
    ├── app.module.ts          # connects to MongoDB, imports CourseModule
    └── course/
        ├── course.schema.ts
        ├── course.controller.ts
        ├── course.service.ts
        └── course.module.ts
```

Schema (course.schema.ts)

```
@Schema()
export class Course {
  @Prop({ required: true, unique: true }) code!: string;
  @Prop({ required: true })               name!: string;
  @Prop({ required: true })               instructor!: string;
  @Prop({ required: true })               creditHours!: number;
  @Prop()                                 department!: string;
  @Prop()                                 capacity!: number;
  @Prop()                                 schedule!: string;
}
```

Endpoints

Method	Path	Description
POST	/courses	Create a new course

Method	Path	Description
GET	/courses	List all courses
GET	/courses/:id	Get one course by Mongo _id
PUT	/courses/:id	Update a course
DELETE	/courses/:id	Delete a course

5.3 Registration Service

Responsibility

Acts as the bridge between students and courses. Each registration document links a studentId to a courseId for a given semester and tracks its status (e.g. active, dropped). Because this is the relational glue of the system, it also exposes look-up endpoints by student and by course.

Port

localhost:3003

Folder structure

```
registration-service/
└── src/
    ├── main.ts                      # boots NestJS on port 3003
    ├── app.module.ts                # connects to MongoDB
    └── registration/
        ├── registration.schema.ts
        ├── registration.controller.ts
        ├── registration.service.ts
        └── registration.module.ts
```

Schema (registration.schema.ts)

```
@Schema({ timestamps: true })
export class Registration {
  @Prop({ required: true })       studentId!: string;
  @Prop({ required: true })       courseId!: string;
  @Prop()                         semester!: string;
  @Prop({ default: 'active' })    status!: string;
}
```

The timestamps option automatically adds createdAt and updatedAt fields, which is useful for auditing the registration history.

Endpoints

Method	Path	Description
POST	/registrations	Register a student in a course

Method	Path	Description
GET	/registrations	List all registrations
GET	/registrations/:id	Get one registration
GET	/registrations/student/:studentId	All registrations for a student
GET	/registrations/course/:courseId	All registrations for a course
PUT	/registrations/:id	Update a registration
DELETE	/registrations/:id	Delete a registration

6. API Gateway

The API Gateway is a fourth, dedicated NestJS application whose only job is to expose a single, stable entry point for the React frontend and to forward each incoming request to the right downstream microservice.

Why a gateway?

- Hides internal topology — the frontend doesn't need to know that students live on port 3001 and courses on 3002.
- Centralises cross-cutting concerns — CORS, logging, future authentication and rate-limiting all happen in one place.
- Stable contract — even if a microservice is renamed, moved or split in two, the frontend's URLs stay the same.

Folder structure

```
api-gateway/
└── src/
    ├── main.ts                      # boots NestJS on port 3000 + enables CORS
    ├── app.module.ts                # imports GatewayModule
    └── gateway/
        ├── gateway.controller.ts    # 15 endpoints — full pass-through
        ├── gateway.service.ts       # Axios-based forwarders
        └── gateway.module.ts
```

Forwarding logic (gateway.service.ts)

```
@Injectable()
export class GatewayService {
  private studentServiceUrl       = 'http://localhost:3001';
  private courseServiceUrl        = 'http://localhost:3002';
  private registrationServiceUrl  = 'http://localhost:3003';
```

```
  async forwardToStudent(method: string, path: string, data?: any) {
    const response = await axios({ method, url: `${this.studentServiceUrl}${path}`, data
});
    return response.data;
  }
  // forwardToCourse / forwardToRegistration follow the same pattern
}
```

Gateway endpoints

The gateway re-exposes every microservice endpoint at the same path, so the React app only ever talks to localhost:3000:

From frontend	Forwarded to
POST /students	POST localhost:3001/students
GET /students	GET localhost:3001/students
GET /students/:id	GET localhost:3001/students/:id
PUT /students/:id	PUT localhost:3001/students/:id
DELETE /students/:id	DELETE localhost:3001/students/:id
POST /courses	POST localhost:3002/courses
GET /courses	GET localhost:3002/courses
GET /courses/:id	GET localhost:3002/courses/:id
PUT /courses/:id	PUT localhost:3002/courses/:id
DELETE /courses/:id	DELETE localhost:3002/courses/:id
POST /registrations	POST localhost:3003/registrations
GET /registrations	GET localhost:3003/registrations
GET /registrations/:id	GET localhost:3003/registrations/:id
PUT /registrations/:id	PUT localhost:3003/registrations/:id
DELETE /registrations/:id	DELETE localhost:3003/registrations/:id

CORS

In main.ts the gateway enables CORS for all origins (origin: '*'), which lets the React dev server on a different port communicate freely with the gateway during development:

```
const app = await NestFactory.create(AppModule);
app.enableCors({ origin: '*' });
```

```
await app.listen(3000);
```

7. Database Design

Choice of database — and why

All three microservices use MongoDB Atlas, a managed cloud-hosted NoSQL document database. The choice was driven by several factors:

- **Document model fits the data —** Students, courses and registrations are naturally self-contained JSON-shaped records. Storing them as BSON documents removes the impedance mismatch between code and storage.
- **Flexible schema —** During a 3-person student project, schema evolves quickly. MongoDB lets us add fields (e.g. schedule, capacity) without writing migrations.
- **First-class NestJS support —** The @nestjs/mongoose package gives us @Schema(), @Prop() and InjectModel() decorators, matching the dependency-injection style the rest of NestJS uses.
- **Database-per-service made simple —** Atlas lets us point each microservice at its own connection string with zero infrastructure to manage.
- **Cloud hosting —** Atlas means there is no local mongod to install on each teammate's laptop; the same connection string works from anywhere.

> **Multi-database note**
>
> The rubric notes that using multiple database types is appreciated. Our current implementation uses MongoDB across all three services to keep the team focused on getting the microservices contract right. A natural future extension would be to back the Registration Service with a relational store (e.g. PostgreSQL) — registrations are a join table by nature — while keeping Students and Courses as document stores.

Collections

Three collections live inside the shared MongoDB Atlas cluster. Each is owned by exactly one service:

Collection	Owner service	Document shape
students	Student Service	{ _id, name, email, department, level }
courses	Course Service	{ _id, code, name, instructor, creditHours, department, capacity, schedule }
registrations	Registration Service	{ _id, studentId, courseId, semester, status, createdAt, updatedAt }

Schema definitions

students collection

```
{
  _id:        ObjectId,
  name:       String,    // required
  email:      String,    // required, unique
  department: String,    // required
  level:      String     // required, e.g. "3"
}
```

courses collection

```
{
  _id:         ObjectId,
  code:        String,   // required, unique, e.g. "CS101"
  name:        String,   // required, e.g. "Introduction to Programming"
  instructor:  String,   // required
  creditHours: Number,   // required
  department:  String,
  capacity:    Number,
  schedule:    String    // e.g. "Sunday 10:00 AM"
}
```

registrations collection

```
{
  _id:        ObjectId,
  studentId:  String,    // required, references students._id
  courseId:   String,    // required, references courses._id
  semester:   String,    // e.g. "Spring 2026"
  status:     String,    // default "active"
  createdAt:  Date,      // auto-generated by Mongoose timestamps
  updatedAt:  Date       // auto-generated by Mongoose timestamps
}
```

References between collections are stored as plain strings (the string form of the Mongo ObjectId). The Registration Service never embeds the full student or course document — it simply stores their identifiers, leaving each service free to evolve its own schema.

8. Communication Flow Between Microservices

All communication in the system uses synchronous HTTP/JSON. There are two distinct flows:

Flow A — Frontend to backend (via the gateway)

```
React component
    │
    │  axios.get("http://localhost:3000/students")
    ▼
API Gateway (port 3000)
    │
    │  axios.get("http://localhost:3001/students")
    ▼
Student Service (port 3001)
    │
    │  studentModel.find()
    ▼
MongoDB Atlas (students collection)
```

This is the dominant flow — every page in the React UI uses it to fetch and mutate data.

Flow B — Service-to-service (when needed)

Services are intentionally kept independent, but when one service needs to verify data owned by another, it makes a direct HTTP call to that service's REST API. The most representative example in this project is deletion of a student: a student must not be removed while they still have active registrations. The Student Service consults the Registration Service before completing the delete.

```
Student Service                      Registration Service
   │                                          │
   │  GET /registrations/student/:id          │
   ├──────────────────────────────────────────►│
   │                                          │
   │◄──────────────────────────────────────────┤
   │  [...] (array of registrations)          │
   │                                          │
   if (registrations.length > 0) reject delete
   else                          do delete
```

Why synchronous HTTP and not a message broker?

For a CRUD-heavy academic project, synchronous REST keeps things simple, debuggable and directly testable from Postman. A message broker (RabbitMQ, Kafka) would be a natural next step if the system grew to handle asynchronous workflows such as sending confirmation emails or generating reports.

9. API Endpoints (URLs)

Every endpoint below is exposed by the API Gateway at http://localhost:3000. The React frontend uses these URLs exclusively. The downstream microservice URLs are listed for completeness and for direct Postman testing of individual services.

Students

Method	Gateway URL	Direct URL (service)
POST	http://localhost:3000/students	http://localhost:3001/students
GET	http://localhost:3000/students	http://localhost:3001/students
GET	http://localhost:3000/students/:id	http://localhost:3001/students/:id
PUT	http://localhost:3000/students/:id	http://localhost:3001/students/:id
DELETE	http://localhost:3000/students/:id	http://localhost:3001/students/:id

Example body — POST /students:

```
{
  "name":       "Abdelrahman",
  "email":      "abdelrahman@test.com",
  "department": "Software Engineering",
  "level":      "3"
}
```

Courses

Method	Gateway URL	Direct URL (service)
POST	http://localhost:3000/courses	http://localhost:3002/courses
GET	http://localhost:3000/courses	http://localhost:3002/courses
GET	http://localhost:3000/courses/:id	http://localhost:3002/courses/:id
PUT	http://localhost:3000/courses/:id	http://localhost:3002/courses/:id
DELETE	http://localhost:3000/courses/:id	http://localhost:3002/courses/:id

Example body — POST /courses:

```
{
  "code":        "CS101",
  "name":        "Introduction to Programming",
```

```
  "instructor":  "Dr. Ahmed",
  "creditHours": 3,
  "department":  "Software Engineering",
  "capacity":    50,
  "schedule":    "Sunday 10:00 AM"
}
```

Registrations

Method	Gateway URL	Direct URL (service)
POST	http://localhost:3000/registrations	http://localhost:3003/registrations
GET	http://localhost:3000/registrations	http://localhost:3003/registrations
GET	http://localhost:3000/registrations/:id	http://localhost:3003/registrations/:id
GET	http://localhost:3000/registrations/student/:studentId	http://localhost:3003/registrations/student/:studentId
GET	http://localhost:3000/registrations/course/:courseId	http://localhost:3003/registrations/course/:courseId
PUT	http://localhost:3000/registrations/:id	http://localhost:3003/registrations/:id
DELETE	http://localhost:3000/registrations/:id	http://localhost:3003/registrations/:id

Example body — POST /registrations:

```
{
  "studentId": "69b83fda4b947a419a734f8d",
  "courseId":  "69b8404bb945f351a49b13ca",
  "semester":  "Spring 2026",
  "status":    "registered"
}
```

10. Frontend Application

The frontend is a single-page React application (React 19 + React Router 7) that consumes the API Gateway. It is intentionally clean and dashboard-style so that a registrar or administrative user can manage the whole system from one window.

Pages

Route	Component	Purpose
/	DashboardPage	Counts of students, courses & registrations + system overview
/students	StudentsPage	Add, edit, search and delete students
/courses	CoursesPage	Add, edit and delete courses
/registrations	RegistrationsPage	Create, view and remove registrations

Service layer

All API calls are isolated in src/services/ so components never call axios directly. Each file pins to the gateway URL:

```
// src/services/studentService.js
const API_URL = "http://localhost:3000/students";
export const getStudents     = ()       => axios.get(API_URL);
export const getStudentById  = (id)     => axios.get(`${API_URL}/${id}`);
export const createStudent   = (data)   => axios.post(API_URL, data);
export const updateStudent   = (id, d)  => axios.put(`${API_URL}/${id}`, d);
export const deleteStudent   = (id)     => axios.delete(`${API_URL}/${id}`);
```

The same pattern is repeated for courseService.js and registrationService.js.

Component breakdown

- Navbar — top bar with links to Dashboard / Students / Courses / Registrations.
- StudentForm + StudentList — form on the left, searchable list on the right.
- CourseForm + CourseList — same layout for courses.
- RegistrationForm + RegistrationList — registration creation and management.

Dashboard preview

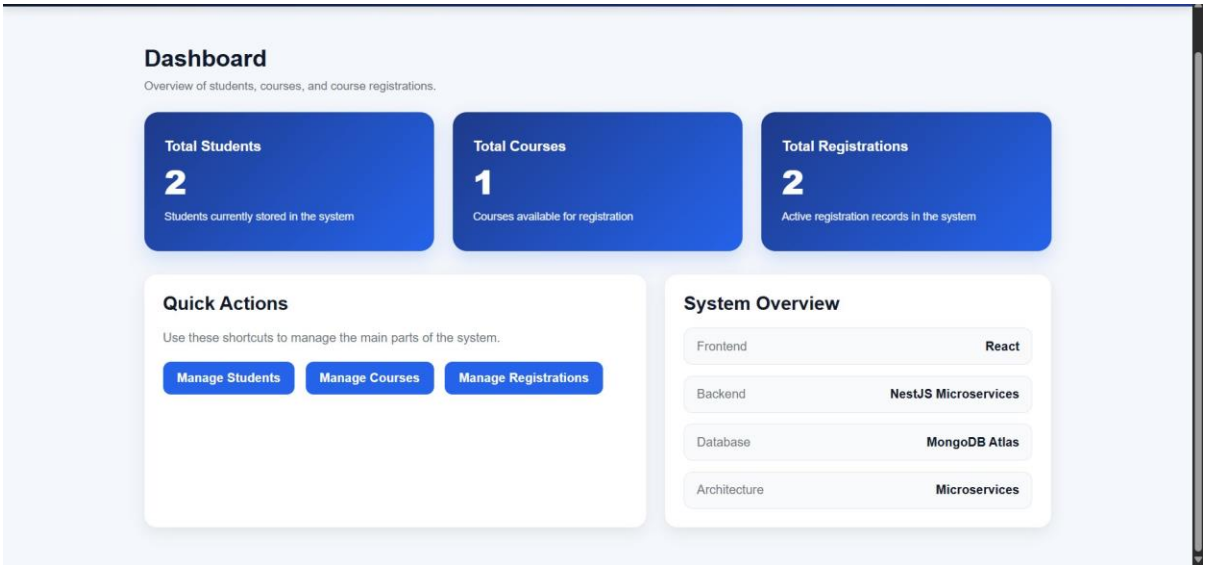

Figure 1 — The Dashboard page, served by the React frontend and populated by live calls to the API Gateway.

11. Testing

All microservices were tested manually with Postman before the React frontend was wired in. The bundled Postman collection (Student Course Registration APIs.postman_collection.json) contains the core requests used to exercise each service end-to-end.

How to test

- Start MongoDB Atlas (cloud — already running) and confirm the connection string in each service's app.module.ts.
- Run the three microservices and the gateway (see Section 12 for the exact commands).
- Import the Postman collection into Postman.
- Fire the requests against http://localhost:3000 to test through the gateway, or against the individual service ports (3001 / 3002 / 3003) to test a service in isolation.

Test scenarios covered

#	Scenario	Expected result
1	POST /students with valid body	201 Created, returns saved student with _id
2	POST /students with duplicate email	MongoDB unique-index error (E11000)
3	GET /students	200 OK, array of all students
4	GET /students/:id with a valid id	200 OK, single student object

#	Scenario	Expected result
5	PUT /students/:id	200 OK, updated student returned
6	DELETE /students/:id	200 OK, student removed
7	POST /courses with valid body	201 Created, returns saved course
8	POST /courses with duplicate code	Unique constraint error on "code"
9	Full CRUD on /courses (GET, PUT, DELETE)	All operations succeed
10	POST /registrations with valid studentId and courseId	201 Created, status defaults to "active"
11	GET /registrations/student/:studentId	200 OK, all registrations for that student
12	GET /registrations/course/:courseId	200 OK, all registrations for that course
13	End-to-end through gateway (http://localhost:3000/...)	Same results as the direct calls in 1–12

Validation strategy

Validation is enforced at three levels:

- Mongoose schema-level — required: true and unique: true on key fields catch missing or duplicate data at write time.
- Frontend form-level — required HTML inputs prevent empty submissions in the React UI.
- Service-level (Registration Service) — uses the existence of studentId and courseId references to keep the data model consistent.

12. Setup & Running the System

Prerequisites

- Node.js 18 or newer
- npm 9+
- A MongoDB Atlas cluster and its connection string
- Postman (for API testing)

Installation

From the project root, install dependencies in each service and in the frontend:

```
cd api-gateway          && npm install
cd ../student-service   && npm install
cd ../course-service    && npm install
cd ../registration-service && npm install
cd ../frontend/frontend-app && npm install
```

Running the backend

Open four terminals and start each NestJS application:

```
# Terminal 1 — Student Service       (port 3001)
cd student-service && npm run start:dev

# Terminal 2 — Course Service        (port 3002)
cd course-service && npm run start:dev

# Terminal 3 — Registration Service  (port 3003)
cd registration-service && npm run start:dev

# Terminal 4 — API Gateway           (port 3000)
cd api-gateway && npm run start:dev
```

Running the frontend

```
cd frontend/frontend-app
npm start
# opens http://localhost:3001 (or next free port) in the browser
```

MongoDB connection

Each service connects to MongoDB Atlas through MongooseModule.forRoot() inside its own app.module.ts. The connection string is currently hard-coded for the project demo; in a production deployment it would be moved into a .env file and loaded with @nestjs/config.

13. Team Responsibilities

The rubric requires exactly one microservice per team member. With three members we have three microservices, plus shared work on the gateway and frontend.

Member	Primary microservice	Shared work
Abdelrahman Hossam	Student Service	API Gateway routing
Kareem Resha	Course Service	MongoDB Atlas setup & schemas
Youssef Khaled	Registration Service	React frontend & integration

Cross-cutting work — the API Gateway, the React UI, the shared MongoDB Atlas cluster and the documentation — was reviewed by all three members so every teammate understands every layer of the system.

14. GitHub Repository

The full source code is hosted on GitHub for version control and team collaboration. The repository contains:

- api-gateway/ — NestJS API Gateway
- student-service/ — NestJS Student microservice
- course-service/ — NestJS Course microservice
- registration-service/ — NestJS Registration microservice
- frontend/frontend-app/ — React frontend
- postman/ — Postman collection for API testing

Repository URL

https://github.com/<your-team>/Student-Course-Registration-System

(replace with the actual repo link before submission)

15. Conclusion

This project delivered a working Student–Course Registration System built end-to-end on a modern, service-oriented stack. Three independent NestJS microservices — Student, Course and Registration — each own their own MongoDB collection, follow the same module/controller/service/schema layout, and expose a clean REST API. A NestJS API Gateway sits in front of them, giving the React frontend a single, stable entry point and hiding the internal topology of the system.

Along the way the team practised the skills called out by the rubric:

- Designing a microservices architecture with clear service boundaries.
- Building reliable REST APIs with NestJS, Mongoose and TypeScript.
- Crafting a responsive React UI that consumes those APIs through a gateway.
- Modelling data in a NoSQL document database and following the database-per-service pattern.
- Writing API documentation and using Postman for systematic testing.
- Collaborating through Git and GitHub with one teammate per microservice.

The result is a system that is easy to reason about, easy to extend (add a Grades service? Drop in a fourth NestJS app and one more route on the gateway), and faithful to the principles taught in Service-Oriented Computing.

— End of Documentation —